

MD5 Hash Function

Practical / Laboratory Program

1. Aim

To implement the MD5 (Message-Digest Algorithm 5) hash function and generate a fixed-length message digest for a given input.

2. Algorithm

1. Start the program.
2. Read the input message from the user.
3. Initialize the MD5 hash object.
4. Encode the input message into bytes.
5. Update the hash object with the input bytes.
6. Compute the MD5 digest.
7. Display the hexadecimal hash value.
8. Stop the program.

3. Program

```
import hashlib

message = input("Enter the message: ")

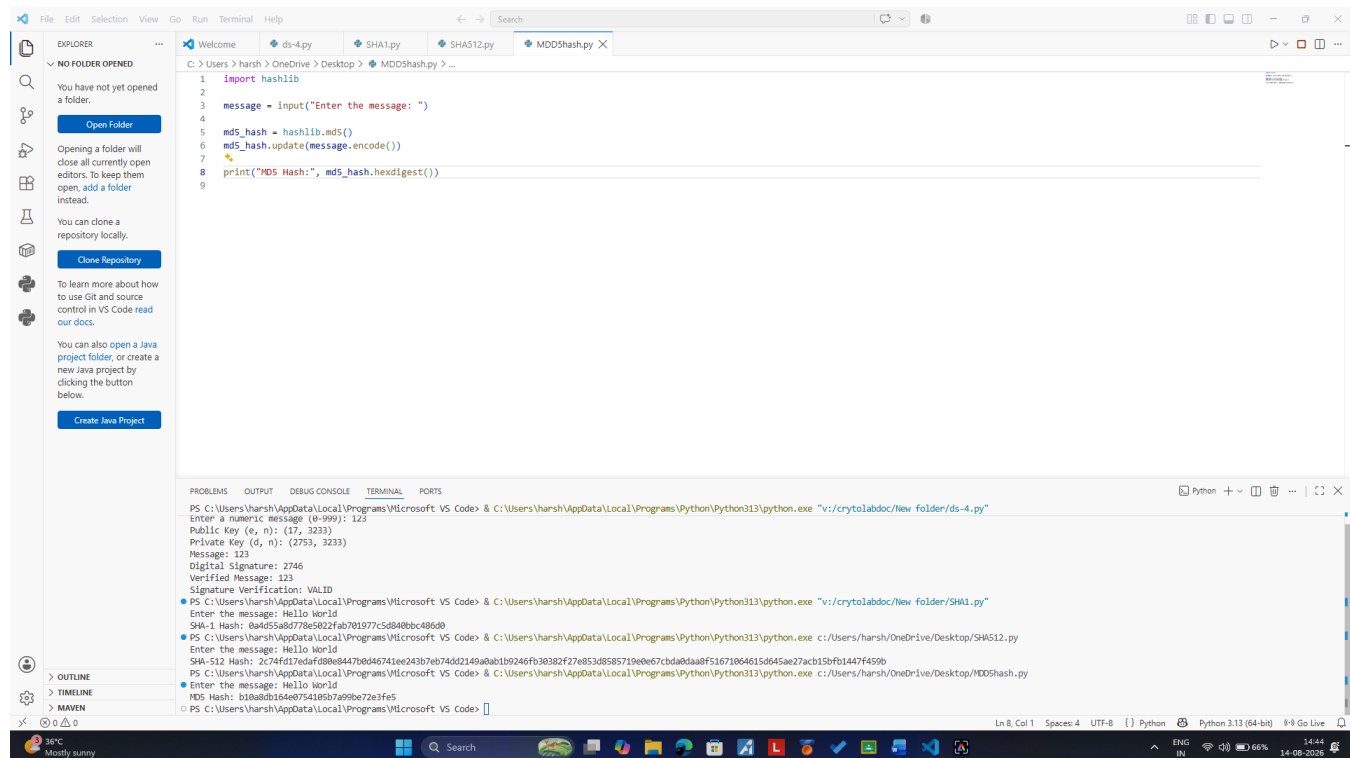
md5_hash = hashlib.md5()
md5_hash.update(message.encode())

print("MD5 Hash:", md5_hash.hexdigest())
```

4. Input

Hello World

5. Output



The screenshot displays the Visual Studio Code interface. The Explorer panel on the left shows a project structure with a file named `MD5hash.py`. The main editor window shows the following Python code:

```
1 import hashlib
2
3 message = input("Enter the message: ")
4
5 md5_hash = hashlib.md5()
6 md5_hash.update(message.encode())
7
8 print("MD5 Hash:", md5_hash.hexdigest())
9
```

The Terminal panel at the bottom shows the execution of the script. It displays the prompt `PS C:\Users\harsh\AppData\Local\Programs\Python\Python313\python.exe` and the command `python.exe "v:/cryptolabdoc/New folder/ds-4.py"`. The output shows the user entering a numeric message (123), the generation of a public key and private key, and the successful verification of the message. The final output is the MD5 hash: `b10aadb164e0754185b7a99be72e3fe5`.

6. Result

Thus, the MD5 hash function was implemented successfully and the hexadecimal message digest was generated.